

Chapter Contents

Chapter 26 - Matrix Solution Methods	3
The Matrix Initial-Value Problem	4
The Goal Of This Block	4
Notation Before We Start	4
Expand A Two-Component System	5
Homogeneous And Forced Parts	6
Block 1 Summary	6
Homogeneous Systems And State Propagation	7
The Goal Of This Block	7
The Homogeneous Formula	7
Verify The Differential Equation	8
The State-Transition Matrix	8
A Nonzero Initial-Time Example	9
A Coupled Nilpotent Example	10
Block 2 Summary	11
Deriving The Forced Solution Formula	12
The Goal Of This Block	12
Why A Constant Vector Is No Longer Enough	12
Differentiate The Substitution	12
Integrate And Apply The Initial State	13
Convert To The Convolution Form	14
Interpret The Two Terms	15
Check The Initial Condition Locally	16
Block 3 Summary	16
Complete Forced-System Calculations	17
The Goal Of This Block	17
A Diagonal Forced System	17
Verify Both Rows Of The System	20
A Forced Triangular System	21
Why The First Component Becomes Quadratic	23
Block 4 Summary	23
Solving A Scalar Equation Through Its State Vector	24
The Goal Of This Block	24
The Initial-Value Problem	24

Calculate The Matrix Exponential	25
Apply The Initial State And Recover Y	26
Verify The Scalar Problem	27
What The Matrix Method Has Added	28
Block 5 Summary	29
Solutions Without Prescribed Initial Conditions	29
The Goal Of This Block	29
The Homogeneous General Solution	29
The Forced General Solution	30
A General Forced Example	31
Verify The General Forced Example	32
Block 6 Summary	33
Choosing A Formula And Avoiding Common Errors	33
The Goal Of This Block	33
The Three Main Forms	33
A Reliable Matrix-Method Workflow	34
Common Mistake: Exponentiating Each Entry	35
Common Mistake: Forgetting The Initial-Time Shift	35
Common Mistake: Using T Inside The Integral	36
Common Mistake: Multiplying In The Wrong Order	36
Common Mistake: Recovering The Wrong Component	36
The Laplace-Domain Connection	37
Block 7 Summary	37
Original Practice Set	38
Problems 1 To 4: Homogeneous Foundations	38
Problems 5 To 8: Forcing And General Solutions	38
Problems 9 To 12: Scalar Recovery And Method Checks	39
Worked Answers: Problems 1 To 4	40
Worked Answers: Problems 5 To 8	43
Worked Answers: Problems 9 To 12	48
Chapter 26 Final Summary	53

Chapter 26 - Matrix Solution Methods

Earlier chapters developed three parts of one larger method:

- Chapter 16 explained how to calculate the matrix exponential e^{At}
- Chapter 17 converted higher-order equations into first-order systems
- Chapter 25 solved coupled systems in the Laplace domain

This chapter joins those parts together in the time domain. We will solve constant-coefficient systems directly with the matrix exponential, derive the forced solution formula step by step, handle initial times other than zero, and recover scalar solutions from state vectors.

The central problem is:

$$\mathbf{x}'(t) = A\mathbf{x}(t) + \mathbf{f}(t), \quad \mathbf{x}(t_0) = \mathbf{x}_0,$$

where A is a constant square matrix.

The Matrix Initial-Value Problem

The Goal Of This Block

The goal is to identify the role of every object in a matrix differential equation and separate the effect of the initial state from the effect of the forcing.

the matrix exponential moves a state through time; an integral adds the contributions made by forcing along the way.

Notation Before We Start

Consider:

$$\mathbf{x}'(t) = A\mathbf{x}(t) + \mathbf{f}(t).$$

Here:

- t is the independent variable
- $\mathbf{x}(t)$ is an n -component state vector
- $\mathbf{x}'(t)$ contains the derivative of every state component
- A is a constant $n \times n$ matrix
- $\mathbf{f}(t)$ is an n -component forcing vector
- t_0 is the time at which the initial state is specified
- $\mathbf{x}_0 = \mathbf{x}(t_0)$ is the initial state
- I is the $n \times n$ identity matrix

The matrix A must have as many rows and columns as $\mathbf{x}$ has components. Then $A\mathbf{x}$ is another n -component vector, so it can be added to $\mathbf{f}(t)$.

Expand A Two-Component System

Let:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix}, \quad \mathbf{f} = \begin{bmatrix} f \\ g \end{bmatrix}.$$

Then:

$$\mathbf{x}' = \begin{bmatrix} x' \\ y' \end{bmatrix}$$

and:

$$\begin{aligned} A\mathbf{x} + \mathbf{f} &= \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} f \\ g \end{bmatrix} \\ &= \begin{bmatrix} ax + by \\ cx + dy \end{bmatrix} + \begin{bmatrix} f \\ g \end{bmatrix} \\ &= \begin{bmatrix} ax + by + f \\ cx + dy + g \end{bmatrix}. \end{aligned}$$

Therefore the matrix equation is exactly the scalar system:

$$\begin{aligned} x' &= ax + by + f(t), \\ y' &= cx + dy + g(t). \end{aligned}$$

The matrix notation compresses the system; it does not change it.

Homogeneous And Forced Parts

If $\mathbf{f}(t) = \mathbf{0}$, the system is homogeneous:

$$\mathbf{x}' = A\mathbf{x}.$$

If $\mathbf{f}(t) \neq \mathbf{0}$, the system is forced or nonhomogeneous:

$$\mathbf{x}' = A\mathbf{x} + \mathbf{f}(t).$$

The complete response naturally divides into:

complete response = response to the initial state + response to the forcing.

That division will become visible in the solution formula.

Block 1 Summary

A constant matrix A controls how the state components interact. The vector $\mathbf{x}_0$ supplies the starting state, while $\mathbf{f}(t)$ supplies external input after the starting time.

Homogeneous Systems And State Propagation

The Goal Of This Block

The goal is to solve a homogeneous matrix initial-value problem and understand why the time difference $t - t_0$ appears.

The Homogeneous Formula

For:

$$\mathbf{x}'(t) = A\mathbf{x}(t), \quad \mathbf{x}(t_0) = \mathbf{x}_0,$$

the solution is:

$$\boxed{\mathbf{x}(t) = e^{A(t-t_0)}\mathbf{x}_0}.$$

The exponent contains elapsed time:

$$t - t_0.$$

At the initial time, no time has elapsed. Substituting $t = t_0$ gives:

$$\begin{aligned} \mathbf{x}(t_0) &= e^{A(t_0-t_0)}\mathbf{x}_0 \\ &= e^{A0}\mathbf{x}_0 \\ &= I\mathbf{x}_0 \\ &= \mathbf{x}_0. \end{aligned}$$

The shift is therefore required when the initial time is not zero.

Verify The Differential Equation

Start from the proposed solution:

$$\mathbf{x}(t) = e^{A(t-t_0)} \mathbf{x}_0.$$

Differentiate it. Since A and $\mathbf{x}_0$ are constant:

$$\begin{aligned} \mathbf{x}'(t) &= \frac{d}{dt} (e^{A(t-t_0)}) \mathbf{x}_0 \\ &= A e^{A(t-t_0)} \mathbf{x}_0. \end{aligned}$$

The expression after A is $\mathbf{x}(t)$, so:

$$\mathbf{x}'(t) = A\mathbf{x}(t).$$

The formula satisfies both parts of the initial-value problem.

The State-Transition Matrix

It is useful to name the matrix that moves a state from t_0 to t :

$$\boxed{\Phi(t, t_0) = e^{A(t-t_0)}}.$$

Then the homogeneous solution becomes:

$$\mathbf{x}(t) = \Phi(t, t_0) \mathbf{x}_0.$$

If $t_0 < t_1 < t_2$, moving first from t_0 to t_1 and then from t_1 to t_2 gives:

$$\begin{aligned} \Phi(t_2, t_1) \Phi(t_1, t_0) &= e^{A(t_2-t_1)} e^{A(t_1-t_0)} \\ &= e^{A[(t_2-t_1)+(t_1-t_0)]} \\ &= e^{A(t_2-t_0)} \\ &= \Phi(t_2, t_0). \end{aligned}$$

The middle time cancels because the two elapsed intervals join into one.

A Nonzero Initial-Time Example

Solve:

$$\mathbf{x}' = \begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix} \mathbf{x}, \quad \mathbf{x}(3) = \begin{bmatrix} 4 \\ -2 \end{bmatrix}.$$

Here:

$$A = \begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix}, \quad t_0 = 3.$$

Because A is diagonal:

$$e^{A(t-3)} = \begin{bmatrix} e^{2(t-3)} & 0 \\ 0 & e^{-(t-3)} \end{bmatrix}.$$

Apply the homogeneous formula:

$$\begin{aligned} \mathbf{x}(t) &= e^{A(t-3)} \mathbf{x}(3) \\ &= \begin{bmatrix} e^{2(t-3)} & 0 \\ 0 & e^{-(t-3)} \end{bmatrix} \begin{bmatrix} 4 \\ -2 \end{bmatrix} \\ &= \begin{bmatrix} 4e^{2(t-3)} \\ -2e^{-(t-3)} \end{bmatrix}. \end{aligned}$$

Therefore:

$$\boxed{x_1(t) = 4e^{2(t-3)}, \quad x_2(t) = -2e^{-(t-3)}}.$$

Set $t = 3$ to check the starting values:

$$x_1(3) = 4e^0 = 4, \quad x_2(3) = -2e^0 = -2.$$

A Coupled Nilpotent Example

Solve:

$$\mathbf{x}' = \begin{bmatrix} -1 & 1 \\ 0 & -1 \end{bmatrix} \mathbf{x}, \quad \mathbf{x}(0) = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

Split the matrix as:

$$A = -I + N, \quad N = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

Calculate N^2 :

$$N^2 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

Since $N^2 = 0$, its exponential stops after the linear term:

$$e^{Nt} = I + tN = \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix}.$$

The matrices $-I$ and N commute, so:

$$\begin{aligned} e^{At} &= e^{(-I+N)t} \\ &= e^{-tI} e^{Nt} \\ &= e^{-t} \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix}. \end{aligned}$$

Using the calculated exponential, apply $\mathbf{x}(t) = e^{At}\mathbf{x}(0)$:

$$\begin{aligned}\mathbf{x}(t) &= e^{At}\mathbf{x}(0) \\ &= e^{-t} \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} \\ &= e^{-t} \begin{bmatrix} 1 + 2t \\ 2 \end{bmatrix}.\end{aligned}$$

Thus:

$$\boxed{x_1(t) = (1 + 2t)e^{-t}, \quad x_2(t) = 2e^{-t}}.$$

The factor t appears because the off-diagonal coupling transfers the second component into the first.

Block 2 Summary

For a homogeneous constant system, $e^{A(t-t_0)}$ carries the initial state forward by the elapsed time. Calculate the exponential first, then multiply it by the complete initial vector.

Deriving The Forced Solution Formula

The Goal Of This Block

The goal is to derive the forced solution rather than present its integral as an unexplained rule.

Why A Constant Vector Is No Longer Enough

For the homogeneous system, a solution has the form:

$$\mathbf{x}(t) = e^{At}\mathbf{c},$$

where $\mathbf{c}$ is constant.

For the forced system:

$$\mathbf{x}' = A\mathbf{x} + \mathbf{f}(t),$$

the forcing continually changes the response. We therefore replace the constant vector by an unknown vector function:

$$\boxed{\mathbf{x}(t) = e^{At}\mathbf{u}(t)}.$$

This is the matrix version of variation of parameters.

Differentiate The Substitution

For $\mathbf{x} = e^{At}\mathbf{u}$, both factors depend on t . Use the product rule:

$$\begin{aligned}\mathbf{x}' &= (e^{At})'\mathbf{u} + e^{At}\mathbf{u}' \\ &= Ae^{At}\mathbf{u} + e^{At}\mathbf{u}'.\end{aligned}$$

The original differential equation is:

$$\mathbf{x}' = A\mathbf{x} + \mathbf{f}(t).$$

Substitute $\mathbf{x} = e^{At}\mathbf{u}$ into its right side:

$$A\mathbf{x} + \mathbf{f}(t) = Ae^{At}\mathbf{u} + \mathbf{f}(t).$$

Equate the two expressions for $\mathbf{x}'$:

$$Ae^{At}\mathbf{u} + e^{At}\mathbf{u}' = Ae^{At}\mathbf{u} + \mathbf{f}(t).$$

Subtract $Ae^{At}\mathbf{u}$ from both sides:

$$e^{At}\mathbf{u}' = \mathbf{f}(t).$$

Multiply on the left by e^{-At} :

$$\begin{aligned} e^{-At}e^{At}\mathbf{u}' &= e^{-At}\mathbf{f}(t) \\ I\mathbf{u}' &= e^{-At}\mathbf{f}(t). \end{aligned}$$

Therefore:

$$\boxed{\mathbf{u}'(t) = e^{-At}\mathbf{f}(t)}.$$

Integrate And Apply The Initial State

Integrate $\mathbf{u}'(s)$ from t_0 to t :

$$\mathbf{u}(t) - \mathbf{u}(t_0) = \int_{t_0}^t e^{-As}\mathbf{f}(s) ds.$$

Hence:

$$\mathbf{u}(t) = \mathbf{u}(t_0) + \int_{t_0}^t e^{-As} \mathbf{f}(s) ds.$$

We still need $\mathbf{u}(t_0)$. From $\mathbf{x}(t) = e^{At} \mathbf{u}(t)$:

$$\mathbf{x}_0 = e^{At_0} \mathbf{u}(t_0).$$

Multiply by e^{-At_0} on the left:

$$\begin{aligned} e^{-At_0} \mathbf{x}_0 &= e^{-At_0} e^{At_0} \mathbf{u}(t_0) \\ &= I \mathbf{u}(t_0) \\ &= \mathbf{u}(t_0). \end{aligned}$$

Therefore:

$$\mathbf{u}(t) = e^{-At_0} \mathbf{x}_0 + \int_{t_0}^t e^{-As} \mathbf{f}(s) ds.$$

Return to $\mathbf{x}(t) = e^{At} \mathbf{u}(t)$:

$$\boxed{\mathbf{x}(t) = e^{At} \left[e^{-At_0} \mathbf{x}_0 + \int_{t_0}^t e^{-As} \mathbf{f}(s) ds \right]}.$$

This is the factored form of the forced solution.

Convert To The Convolution Form

Distribute e^{At} across the two terms:

$$\mathbf{x}(t) = e^{At} e^{-At_0} \mathbf{x}_0 + e^{At} \int_{t_0}^t e^{-As} \mathbf{f}(s) ds.$$

Because all exponentials here contain the same matrix A :

$$e^{At}e^{-At_0} = e^{A(t-t_0)}.$$

The matrix e^{At} does not depend on the integration variable s , so move it inside the integral:

$$e^{At} \int_{t_0}^t e^{-As} \mathbf{f}(s) ds = \int_{t_0}^t e^{At} e^{-As} \mathbf{f}(s) ds.$$

Combine the matrix exponentials inside the integrand:

$$e^{At}e^{-As} = e^{A(t-s)}.$$

The result is:

$$\mathbf{x}(t) = e^{A(t-t_0)} \mathbf{x}_0 + \int_{t_0}^t e^{A(t-s)} \mathbf{f}(s) ds.$$

This is the convolution form. The variable s is a temporary integration variable; after integration, the answer is a function of t .

Interpret The Two Terms

The first term is:

$$e^{A(t-t_0)} \mathbf{x}_0.$$

It is the response that would remain if the forcing were removed.

The second term is:

$$\int_{t_0}^t e^{A(t-s)} \mathbf{f}(s) ds.$$

At time s , the forcing contributes a small state change $\mathbf{f}(s) ds$. The factor $e^{A(t-s)}$ then carries that contribution from time s to the observation time t .¹

In everyday terms:

the solution remembers both where the system started and everything that was added after it started.

Check The Initial Condition Locally

Set $t = t_0$ in the convolution form. The transition interval becomes zero and the integral receives equal limits:

$$\begin{aligned} \mathbf{x}(t_0) &= e^{A(t_0-t_0)} \mathbf{x}_0 + \int_{t_0}^{t_0} e^{A(t_0-s)} \mathbf{f}(s) ds \\ &= I \mathbf{x}_0 + \mathbf{0} \\ &= \mathbf{x}_0. \end{aligned}$$

The integral vanishes because its upper and lower limits are equal.

Block 3 Summary

¹ The integral is a continuous sum of contributions made at different times. Earlier contributions evolve for longer because their elapsed time $t - s$ is larger.

Replacing the constant vector in $e^{At}\mathbf{c}$ by a vector function produces the forced solution. The two useful forms are:

$$\mathbf{x}(t) = e^{At} \left[e^{-At_0} \mathbf{x}_0 + \int_{t_0}^t e^{-As} \mathbf{f}(s) ds \right]$$

and:

$$\mathbf{x}(t) = e^{A(t-t_0)} \mathbf{x}_0 + \int_{t_0}^t e^{A(t-s)} \mathbf{f}(s) ds.$$

Complete Forced-System Calculations

The Goal Of This Block

The goal is to calculate the forcing integral without concealing the matrix substitution or component integrals.

A Diagonal Forced System

Solve:

$$\mathbf{x}' = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 1 \\ e^{-t} \end{bmatrix}, \quad \mathbf{x}(0) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

For:

$$A = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix},$$

the transition matrix over an interval $t - s$ is:

$$e^{A(t-s)} = \begin{bmatrix} e^{-(t-s)} & 0 \\ 0 & e^{-2(t-s)} \end{bmatrix}.$$

Use the convolution formula with $t_0 = 0$:

$$\mathbf{x}(t) = e^{At}\mathbf{x}(0) + \int_0^t e^{A(t-s)}\mathbf{f}(s) ds.$$

First calculate the initial-state response:

$$\begin{aligned} e^{At}\mathbf{x}(0) &= \begin{bmatrix} e^{-t} & 0 \\ 0 & e^{-2t} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} 0 \\ e^{-2t} \end{bmatrix}. \end{aligned}$$

Next multiply the transition matrix by the forcing at time s :

$$\begin{aligned} e^{A(t-s)}\mathbf{f}(s) &= \begin{bmatrix} e^{-(t-s)} & 0 \\ 0 & e^{-2(t-s)} \end{bmatrix} \begin{bmatrix} 1 \\ e^{-s} \end{bmatrix} \\ &= \begin{bmatrix} e^{-(t-s)} \\ e^{-2(t-s)}e^{-s} \end{bmatrix}. \end{aligned}$$

Integrate the first component:

$$\begin{aligned} \int_0^t e^{-(t-s)} ds &= e^{-t} \int_0^t e^s ds \\ &= e^{-t} [e^s]_0^t \\ &= e^{-t}(e^t - 1) \\ &= 1 - e^{-t}. \end{aligned}$$

For the second component, first combine the exponents:

$$\begin{aligned} e^{-2(t-s)}e^{-s} &= e^{-2t+2s-s} \\ &= e^{-2t+s} \\ &= e^{-2t}e^s. \end{aligned}$$

Now integrate:

$$\begin{aligned} \int_0^t e^{-2(t-s)}e^{-s} ds &= e^{-2t} \int_0^t e^s ds \\ &= e^{-2t}(e^t - 1) \\ &= e^{-t} - e^{-2t}. \end{aligned}$$

Therefore the forcing response is:

$$\int_0^t e^{A(t-s)}\mathbf{f}(s) ds = \begin{bmatrix} 1 - e^{-t} \\ e^{-t} - e^{-2t} \end{bmatrix}.$$

Add the initial-state and forcing responses:

$$\begin{aligned} \mathbf{x}(t) &= \begin{bmatrix} 0 \\ e^{-2t} \end{bmatrix} + \begin{bmatrix} 1 - e^{-t} \\ e^{-t} - e^{-2t} \end{bmatrix} \\ &= \begin{bmatrix} 1 - e^{-t} \\ e^{-t} \end{bmatrix}. \end{aligned}$$

Thus:

$$\boxed{x_1(t) = 1 - e^{-t}, \quad x_2(t) = e^{-t}}.$$

Verify Both Rows Of The System

The proposed solution is:

$$x_1 = 1 - e^{-t}, \quad x_2 = e^{-t}.$$

Differentiate:

$$x'_1 = e^{-t}, \quad x'_2 = -e^{-t}.$$

The first differential equation is $x'_1 = -x_1 + 1$. Its right side is:

$$\begin{aligned} -x_1 + 1 &= -(1 - e^{-t}) + 1 \\ &= e^{-t} \\ &= x'_1. \end{aligned}$$

The second equation is $x'_2 = -2x_2 + e^{-t}$. Its right side is:

$$\begin{aligned} -2x_2 + e^{-t} &= -2e^{-t} + e^{-t} \\ &= -e^{-t} \\ &= x'_2. \end{aligned}$$

Finally:

$$x_1(0) = 0, \quad x_2(0) = 1.$$

Both equations and both initial values are satisfied.

A Forced Triangular System

Solve:

$$\mathbf{x}' = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 2 \end{bmatrix}, \quad \mathbf{x}(0) = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Let:

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

Since $A^2 = 0$:

$$e^{Ar} = I + rA = \begin{bmatrix} 1 & r \\ 0 & 1 \end{bmatrix}.$$

Use $r = t - s$ for the forcing term:

$$e^{A(t-s)} = \begin{bmatrix} 1 & t-s \\ 0 & 1 \end{bmatrix}.$$

The initial-state response is:

$$\begin{aligned} e^{At}\mathbf{x}(0) &= \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} \\ &= \begin{bmatrix} 1-t \\ -1 \end{bmatrix}. \end{aligned}$$

Now calculate the forcing integrand:

$$\begin{aligned} e^{A(t-s)}\mathbf{f}(s) &= \begin{bmatrix} 1 & t-s \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \end{bmatrix} \\ &= \begin{bmatrix} 2(t-s) \\ 2 \end{bmatrix}. \end{aligned}$$

Integrate each component:

$$\begin{aligned}\int_0^t 2(t-s) \, ds &= [2ts - s^2]_0^t \\ &= 2t^2 - t^2 \\ &= t^2,\end{aligned}$$

and:

$$\int_0^t 2 \, ds = 2t.$$

Therefore:

$$\int_0^t e^{A(t-s)} \mathbf{f}(s) \, ds = \begin{bmatrix} t^2 \\ 2t \end{bmatrix}.$$

Add both responses:

$$\begin{aligned}\mathbf{x}(t) &= \begin{bmatrix} 1-t \\ -1 \end{bmatrix} + \begin{bmatrix} t^2 \\ 2t \end{bmatrix} \\ &= \begin{bmatrix} 1-t+t^2 \\ -1+2t \end{bmatrix}.\end{aligned}$$

Thus:

$$\boxed{x_1(t) = 1 - t + t^2, \quad x_2(t) = -1 + 2t}.$$

Why The First Component Becomes Quadratic

Expanding the system gives:

$$x_1' = x_2, \quad x_2' = 2.$$

The constant forcing makes x_2 linear. Because $x_1' = x_2$, integrating that linear function makes x_1 quadratic.

The same mechanism appeared inside the matrix integral as the factor $t - s$.

Block 4 Summary

In a forced calculation, keep three stages visible:

1. calculate the initial-state response
2. multiply $e^{A(t-s)}$ by $\mathbf{f}(s)$ before integrating
3. add the two response vectors component by component

This prevents the transition matrix, forcing vector, and integration variable from becoming mixed together.

Solving A Scalar Equation Through Its State Vector

The Goal Of This Block

The goal is to complete the entire chain from a scalar differential equation to a matrix solution and then back to the original unknown function.

The Initial-Value Problem

Solve:

$$y'' + 3y' + 2y = 0, \quad y(0) = 1, \quad y'(0) = 0.$$

Define the state components:

$$x_1 = y, \quad x_2 = y'.$$

Differentiate the first state definition:

$$x'_1 = y' = x_2.$$

Isolate y'' in the original equation:

$$y'' = -2y - 3y'.$$

Replace y by x_1 and y' by x_2 :

$$x'_2 = -2x_1 - 3x_2.$$

Stack the two state equations:

$$\begin{bmatrix} x_1' \\ x_2' \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}.$$

Thus:

$$\mathbf{x}' = A\mathbf{x}, \quad A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}.$$

Convert the initial data in the declared state order:

$$\mathbf{x}(0) = \begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix} = \begin{bmatrix} y(0) \\ y'(0) \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Calculate The Matrix Exponential

The characteristic polynomial of A is:

$$\begin{aligned} \det(\lambda I - A) &= \det \begin{bmatrix} \lambda & -1 \\ 2 & \lambda + 3 \end{bmatrix} \\ &= \lambda(\lambda + 3) - (-1)(2) \\ &= \lambda^2 + 3\lambda + 2 \\ &= (\lambda + 1)(\lambda + 2). \end{aligned}$$

The distinct eigenvalues are -1 and -2 . By the two-eigenvalue interpolation formula, write:

$$e^{At} = \alpha(t)I + \beta(t)A.$$

At $\lambda = -1$ and $\lambda = -2$, the scalar polynomial $\alpha + \beta\lambda$ must match $e^{\lambda t}$:

$$\begin{aligned}\alpha - \beta &= e^{-t}, \\ \alpha - 2\beta &= e^{-2t}.\end{aligned}$$

Subtract the second equation from the first:

$$\beta = e^{-t} - e^{-2t}.$$

Substitute this into $\alpha - \beta = e^{-t}$:

$$\begin{aligned}\alpha - (e^{-t} - e^{-2t}) &= e^{-t}, \\ \alpha &= 2e^{-t} - e^{-2t}.\end{aligned}$$

Therefore:

$$e^{At} = (2e^{-t} - e^{-2t})I + (e^{-t} - e^{-2t})A.$$

Insert I and A :

$$\begin{aligned}e^{At} &= (2e^{-t} - e^{-2t}) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + (e^{-t} - e^{-2t}) \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \\ &= \begin{bmatrix} 2e^{-t} - e^{-2t} & e^{-t} - e^{-2t} \\ -2e^{-t} + 2e^{-2t} & -e^{-t} + 2e^{-2t} \end{bmatrix}.\end{aligned}$$

Apply The Initial State And Recover Y

Use:

$$\mathbf{x}(t) = e^{At}\mathbf{x}(0).$$

Substitute the calculated matrix and initial state:

$$\begin{aligned}\mathbf{x}(t) &= \begin{bmatrix} 2e^{-t} - e^{-2t} & e^{-t} - e^{-2t} \\ -2e^{-t} + 2e^{-2t} & -e^{-t} + 2e^{-2t} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} 2e^{-t} - e^{-2t} \\ -2e^{-t} + 2e^{-2t} \end{bmatrix}.\end{aligned}$$

Recall the state definitions:

$$x_1 = y, \quad x_2 = y'.$$

The original unknown is the first component, so:

$$\boxed{y(t) = 2e^{-t} - e^{-2t}}.$$

The second component confirms its derivative:

$$\begin{aligned}y'(t) &= \frac{d}{dt}(2e^{-t} - e^{-2t}) \\ &= -2e^{-t} + 2e^{-2t} \\ &= x_2(t).\end{aligned}$$

Verify The Scalar Problem

From:

$$y = 2e^{-t} - e^{-2t},$$

differentiate twice:

$$\begin{aligned}y' &= -2e^{-t} + 2e^{-2t}, \\y'' &= 2e^{-t} - 4e^{-2t}.\end{aligned}$$

Substitute into the left side of the scalar equation:

$$\begin{aligned}y'' + 3y' + 2y &= (2e^{-t} - 4e^{-2t}) + 3(-2e^{-t} + 2e^{-2t}) + 2(2e^{-t} - e^{-2t}) \\&= (2 - 6 + 4)e^{-t} + (-4 + 6 - 2)e^{-2t} \\&= 0.\end{aligned}$$

Check the initial values:

$$y(0) = 2 - 1 = 1,$$

and:

$$y'(0) = -2 + 2 = 0.$$

The recovered first component solves the original initial-value problem.

What The Matrix Method Has Added

For this second-order equation, a characteristic-polynomial method would be shorter. The matrix route becomes especially useful when:

- several equations are coupled
- the state vector has a physical interpretation
- one matrix exponential will be reused for many initial states
- forcing must be added to a system
- a numerical or computational state-space method will follow

The practical meaning:

the matrix method solves every state component together and preserves the relationships between them.

Block 5 Summary

To solve a scalar equation by matrix methods, declare the state order, build the first-order system, convert the initial data in that same order, solve the matrix problem, and explicitly recover the original unknown from its state component.

Solutions Without Prescribed Initial Conditions

The Goal Of This Block

The goal is to write a general solution when no initial state has been given and understand where the arbitrary constants belong.

The Homogeneous General Solution

For:

$$\mathbf{x}' = A\mathbf{x},$$

with no initial condition, write:

$$\boxed{\mathbf{x}(t) = e^{At}\mathbf{c}},$$

where:

$$\mathbf{c} = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}$$

is an arbitrary constant vector.

There are n independent constants because an n -component first-order system requires n scalar initial values to select one solution.

The Forced General Solution

The variation-of-parameters derivation gave:

$$\mathbf{u}'(t) = e^{-At}\mathbf{f}(t).$$

Without initial data, integrate indefinitely:

$$\mathbf{u}(t) = \mathbf{c} + \int e^{-As}\mathbf{f}(s) ds.$$

Substitute this into $\mathbf{x} = e^{At}\mathbf{u}$:

$$\boxed{\mathbf{x}(t) = e^{At} \left[\mathbf{c} + \int e^{-As}\mathbf{f}(s) ds \right]}.$$

The integral means any one vector antiderivative. Constants produced by that antiderivative can be absorbed into the already arbitrary vector $\mathbf{c}$.

For cleaner definite integrals, choose any fixed reference time τ and write:

$$\boxed{\mathbf{x}(t) = e^{A(t-\tau)}\mathbf{c} + \int_{\tau}^t e^{A(t-s)}\mathbf{f}(s) ds}.$$

Here $\mathbf{c}$ represents the arbitrary state at $t = \tau$.

A General Forced Example

Find the general solution of:

$$\mathbf{x}' = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \mathbf{x} + \begin{bmatrix} e^t \\ 1 \end{bmatrix}.$$

Let:

$$A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

Then:

$$e^{At} = \begin{bmatrix} e^t & 0 \\ 0 & e^{-t} \end{bmatrix}, \quad e^{-At} = \begin{bmatrix} e^{-t} & 0 \\ 0 & e^t \end{bmatrix}.$$

Multiply e^{-At} by the forcing:

$$\begin{aligned} e^{-At} \mathbf{f}(t) &= \begin{bmatrix} e^{-t} & 0 \\ 0 & e^t \end{bmatrix} \begin{bmatrix} e^t \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ e^t \end{bmatrix}. \end{aligned}$$

Integrate component by component, choosing one antiderivative with no added constants:

$$\int e^{-At} \mathbf{f}(t) dt = \begin{bmatrix} t \\ e^t \end{bmatrix}.$$

Insert the arbitrary vector:

$$\mathbf{c} = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}.$$

Now multiply by e^{At} :

$$\begin{aligned} \mathbf{x}(t) &= e^{At} \left[\mathbf{c} + \begin{bmatrix} t \\ e^t \end{bmatrix} \right] \\ &= \begin{bmatrix} e^t & 0 \\ 0 & e^{-t} \end{bmatrix} \begin{bmatrix} c_1 + t \\ c_2 + e^t \end{bmatrix} \\ &= \begin{bmatrix} e^t(c_1 + t) \\ c_2 e^{-t} + 1 \end{bmatrix}. \end{aligned}$$

Therefore:

$$\boxed{x_1(t) = c_1 e^t + t e^t, \quad x_2(t) = c_2 e^{-t} + 1}.$$

The term $t e^t$ appears because the first forcing e^t has the same exponential form as the first homogeneous mode. This is the matrix version of resonance in undetermined coefficients.

Verify The General Forced Example

The scalar rows are:

$$x_1' = x_1 + e^t, \quad x_2' = -x_2 + 1.$$

For $x_1 = c_1 e^t + t e^t$, use the product rule:

$$\begin{aligned} x_1' &= c_1 e^t + (t e^t)' \\ &= c_1 e^t + (e^t + t e^t) \\ &= (c_1 e^t + t e^t) + e^t \\ &= x_1 + e^t. \end{aligned}$$

For $x_2 = c_2 e^{-t} + 1$:

$$\begin{aligned} x_2' &= -c_2 e^{-t}, \\ -x_2 + 1 &= -(c_2 e^{-t} + 1) + 1 \\ &= -c_2 e^{-t} \\ &= x_2'. \end{aligned}$$

Both arbitrary constants remain free because no initial values were supplied.

Block 6 Summary

Without initial conditions, retain an arbitrary constant vector. For a homogeneous system use $e^{At}\mathbf{c}$; for a forced system add any one particular response obtained from the matrix integral.

Choosing A Formula And Avoiding Common Errors

The Goal Of This Block

The goal is to choose an efficient form of the solution and keep the matrix algebra logically checkable.

The Three Main Forms

For a homogeneous IVP:

$$\mathbf{x}(t) = e^{A(t-t_0)} \mathbf{x}_0.$$

For a forced IVP, the convolution form is:

$$\mathbf{x}(t) = e^{A(t-t_0)} \mathbf{x}_0 + \int_{t_0}^t e^{A(t-s)} \mathbf{f}(s) ds.$$

The equivalent factored form is:

$$\mathbf{x}(t) = e^{At} \left[e^{-At_0} \mathbf{x}_0 + \int_{t_0}^t e^{-As} \mathbf{f}(s) ds \right].$$

Use whichever form produces the cleaner multiplication and integration. The two formulas describe the same solution.

A Reliable Matrix-Method Workflow

Step 1: Put the problem in standard state form. Write:

$$\mathbf{x}' = A\mathbf{x} + \mathbf{f}(t).$$

If the problem began as a higher-order scalar equation, declare every state component before constructing A .

Step 2: Record the complete initial state. Keep the state order visible:

$$\mathbf{x}(t_0) = \mathbf{x}_0.$$

Step 3: Calculate the required matrix exponential. Use diagonal structure, nilpotence, a scalar shift, diagonalization, Cayley–Hamilton reduction, or another justified method.

Step 4: Choose the correct solution formula. Remove the integral only when $\mathbf{f}(t) = \mathbf{0}$.

Step 5: Show matrix-vector products before integration. Calculate:

$$e^{A(t-s)} \mathbf{f}(s)$$

as a vector before integrating its components.

Step 6: Add the two response vectors. Keep the initial-state response and forcing response separate until both are known.

Step 7: Recover the requested variables. If the state was built from $y, y', \dots$, state which component equals y .

Step 8: Verify locally. Check every differential equation and every supplied initial condition.

Common Mistake: Exponentiating Each Entry

In general:

$$e^{At} \neq \begin{bmatrix} e^{a_{11}t} & \dots & e^{a_{1n}t} \\ \vdots & \ddots & \vdots \\ e^{a_{n1}t} & \dots & e^{a_{nn}t} \end{bmatrix}.$$

The entrywise shortcut works for diagonal matrices only because all off-diagonal entries remain zero and the powers stay diagonal.²

Common Mistake: Forgetting The Initial-Time Shift

If $\mathbf{x}(4) = \mathbf{x}_0$, the homogeneous response is:

$$e^{A(t-4)}\mathbf{x}_0,$$

not $e^{At}\mathbf{x}_0$.

The quick check is to substitute $t = 4$. The correct transition matrix becomes $e^{A0} = I$.

² The matrix exponential is defined by $e^{At} = I + At + (At)^2/2! + \dots$. Matrix multiplication inside the powers is what creates coupling between entries.

Common Mistake: Using t Inside The Integral

In:

$$\int_{t_0}^t e^{A(t-s)} \mathbf{f}(s) ds,$$

t is the observation time and s is the integration variable. The forcing must be written as $\mathbf{f}(s)$ inside the integral.

Writing $\mathbf{f}(t)$ inside and treating it as constant with respect to s usually changes the problem.

Common Mistake: Multiplying In The Wrong Order

The forcing contribution is:

$$e^{A(t-s)} \mathbf{f}(s).$$

The matrix is $n \times n$ and the vector is $n \times 1$, so this product is defined. The reversed product $\mathbf{f}(s)e^{A(t-s)}$ is not defined when $\mathbf{f}$ is a column vector.

Matrix multiplication order is part of the formula.

Common Mistake: Recovering The Wrong Component

If:

$$\mathbf{x} = \begin{bmatrix} y \\ y' \end{bmatrix},$$

then the first component is y and the second is y' . If a different state order was declared, the interpretation changes.

Always repeat the state definitions immediately before extracting the final scalar answer.

The Laplace-Domain Connection

For $t_0 = 0$, transform:

$$\mathbf{x}' = A\mathbf{x} + \mathbf{f}(t), \quad \mathbf{x}(0) = \mathbf{x}_0.$$

The transformed equation is:

$$s\mathbf{X}(s) - \mathbf{x}_0 = A\mathbf{X}(s) + \mathbf{F}(s).$$

Move the transform terms to the left:

$$(sI - A)\mathbf{X}(s) = \mathbf{x}_0 + \mathbf{F}(s).$$

Multiply by $(sI - A)^{-1}$:

$$\mathbf{X}(s) = (sI - A)^{-1}\mathbf{x}_0 + (sI - A)^{-1}\mathbf{F}(s).$$

The transform pair:

$$\mathcal{L}\{e^{At}\} = (sI - A)^{-1}$$

turns the first product into the initial-state response and the second product into a convolution. Thus the Laplace and matrix-exponential methods lead to the same time-domain formula.

Block 7 Summary

Choose the formula from the presence or absence of forcing and initial data. Preserve matrix order, distinguish t from the integration variable, and verify the final result in the original representation.

Original Practice Set

Problems 1 To 4: Homogeneous Foundations

Problem 1: Expand A Matrix System. Write the matrix equation as two scalar differential equations:

$$\mathbf{x}' = \begin{bmatrix} -2 & 3 \\ 1 & -4 \end{bmatrix} \mathbf{x} + \begin{bmatrix} t \\ e^{-t} \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} u \\ v \end{bmatrix}.$$

Problem 2: Use A Nonzero Initial Time. Solve:

$$\mathbf{x}' = \begin{bmatrix} -2 & 0 \\ 0 & 3 \end{bmatrix} \mathbf{x}, \quad \mathbf{x}(1) = \begin{bmatrix} 5 \\ -1 \end{bmatrix}.$$

Problem 3: Use A Nilpotent Exponential. Solve:

$$\mathbf{x}' = \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix} \mathbf{x}, \quad \mathbf{x}(0) = \begin{bmatrix} 1 \\ 3 \end{bmatrix}.$$

Problem 4: Check State-Transition Composition. For constant A , verify directly that:

$$\Phi(t_2, t_1)\Phi(t_1, t_0) = \Phi(t_2, t_0),$$

where $\Phi(t, r) = e^{A(t-r)}$.

Problems 5 To 8: Forcing And General Solutions

Problem 5: Separate Two Responses. Solve:

$$\mathbf{x}' = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 2 \\ 0 \end{bmatrix}, \quad \mathbf{x}(0) = \begin{bmatrix} 0 \\ 4 \end{bmatrix}.$$

Problem 6: Integrate A Coupled Forcing. Solve:

$$\mathbf{x}' = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ t \end{bmatrix}, \quad \mathbf{x}(0) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Problem 7: Two Resonant Components. Solve:

$$\mathbf{x}' = \begin{bmatrix} 1 & 0 \\ 0 & -2 \end{bmatrix} \mathbf{x} + \begin{bmatrix} e^t \\ e^{-2t} \end{bmatrix}, \quad \mathbf{x}(0) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Problem 8: No Initial Conditions. Find the general solution of:

$$\mathbf{x}' = \begin{bmatrix} 0 & 0 \\ 0 & -1 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 2t \\ 1 \end{bmatrix}.$$

Problems 9 To 12: Scalar Recovery And Method Checks

Problem 9: Recover An Oscillating Scalar Solution. Convert and solve by matrix methods:

$$y'' + 4y = 0, \quad y(0) = 0, \quad y'(0) = 3.$$

Problem 10: Recover A Forced Scalar Solution. Convert and solve by matrix methods:

$$y'' + y = 1, \quad y(0) = 0, \quad y'(0) = 0.$$

Problem 11: Diagnose Three Formula Errors. Explain the error in each statement.

1. If $\mathbf{x}' = A\mathbf{x}$ and $\mathbf{x}(5) = \mathbf{x}_0$, then $\mathbf{x}(t) = e^{At}\mathbf{x}_0$.
2. The forced response is $\int_0^t \mathbf{f}(s)e^{A(t-s)} ds$ when $\mathbf{f}$ is a column vector.
3. For every square matrix $A = [a_{ij}]$, the entries of e^{At} are $e^{a_{ij}t}$.

Problem 12: Build The Resolvent Equation. For:

$$\mathbf{x}' = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ e^t \end{bmatrix}, \quad \mathbf{x}(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix},$$

write $(sI - A)\mathbf{X} = \mathbf{x}_0 + \mathbf{F}$ explicitly and calculate $\det(sI - A)$. Do not invert the transforms.

Worked Answers: Problems 1 To 4

Answer 1. Use:

$$\mathbf{x} = \begin{bmatrix} u \\ v \end{bmatrix}, \quad \mathbf{x}' = \begin{bmatrix} u' \\ v' \end{bmatrix}.$$

Multiply the matrix by the state vector:

$$\begin{bmatrix} -2 & 3 \\ 1 & -4 \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} -2u + 3v \\ u - 4v \end{bmatrix}.$$

Add the forcing vector component by component:

$$\begin{bmatrix} u' \\ v' \end{bmatrix} = \begin{bmatrix} -2u + 3v + t \\ u - 4v + e^{-t} \end{bmatrix}.$$

Therefore the scalar system is:

$$\begin{aligned} u' &= -2u + 3v + t, \\ v' &= u - 4v + e^{-t}. \end{aligned}$$

Answer 2. Here:

$$t_0 = 1, \quad A = \begin{bmatrix} -2 & 0 \\ 0 & 3 \end{bmatrix}.$$

The homogeneous formula is:

$$\mathbf{x}(t) = e^{A(t-1)}\mathbf{x}(1).$$

Since A is diagonal:

$$e^{A(t-1)} = \begin{bmatrix} e^{-2(t-1)} & 0 \\ 0 & e^{3(t-1)} \end{bmatrix}.$$

Multiply by the initial state:

$$\begin{aligned} \mathbf{x}(t) &= \begin{bmatrix} e^{-2(t-1)} & 0 \\ 0 & e^{3(t-1)} \end{bmatrix} \begin{bmatrix} 5 \\ -1 \end{bmatrix} \\ &= \begin{bmatrix} 5e^{-2(t-1)} \\ -e^{3(t-1)} \end{bmatrix}. \end{aligned}$$

Thus:

$$\boxed{x_1(t) = 5e^{-2(t-1)}, \quad x_2(t) = -e^{3(t-1)}}.$$

At $t = 1$, both exponential factors equal 1, recovering $(5, -1)^T$.

Answer 3. Let:

$$A = \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix}.$$

Calculate the square:

$$A^2 = \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix} = 0.$$

The exponential series therefore truncates:

$$e^{At} = I + tA = \begin{bmatrix} 1 & 2t \\ 0 & 1 \end{bmatrix}.$$

Apply the initial state:

$$\begin{aligned} \mathbf{x}(t) &= \begin{bmatrix} 1 & 2t \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \end{bmatrix} \\ &= \begin{bmatrix} 1 + 6t \\ 3 \end{bmatrix}. \end{aligned}$$

Hence:

$$\boxed{x_1(t) = 1 + 6t, \quad x_2(t) = 3}.$$

Answer 4. Start from the definition:

$$\Phi(t, r) = e^{A(t-r)}.$$

Substitute it into the left side:

$$\Phi(t_2, t_1)\Phi(t_1, t_0) = e^{A(t_2-t_1)}e^{A(t_1-t_0)}.$$

Both exponentials are functions of the same matrix A , so they commute and their exponents combine:

$$\begin{aligned} e^{A(t_2-t_1)}e^{A(t_1-t_0)} &= e^{A[(t_2-t_1)+(t_1-t_0)]} \\ &= e^{A(t_2-t_0)}. \end{aligned}$$

Return to the definition of Φ :

$$\boxed{e^{A(t_2-t_0)} = \Phi(t_2, t_0)}.$$

Worked Answers: Problems 5 To 8

Answer 5. Here $A = -I$, so:

$$e^{A(t-s)} = e^{-(t-s)}I.$$

Use:

$$\mathbf{x}(t) = e^{At}\mathbf{x}(0) + \int_0^t e^{A(t-s)}\mathbf{f}(s) ds.$$

The initial-state response is:

$$\begin{aligned} e^{At}\mathbf{x}(0) &= e^{-t}I \begin{bmatrix} 0 \\ 4 \end{bmatrix} \\ &= \begin{bmatrix} 0 \\ 4e^{-t} \end{bmatrix}. \end{aligned}$$

The forcing integrand is:

$$\begin{aligned} e^{A(t-s)}\mathbf{f}(s) &= e^{-(t-s)}I \begin{bmatrix} 2 \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} 2e^{-(t-s)} \\ 0 \end{bmatrix}. \end{aligned}$$

Integrate its first component:

$$\begin{aligned} \int_0^t 2e^{-(t-s)} ds &= 2e^{-t} \int_0^t e^s ds \\ &= 2e^{-t}(e^t - 1) \\ &= 2(1 - e^{-t}). \end{aligned}$$

Therefore:

$$\begin{aligned} \mathbf{x}(t) &= \begin{bmatrix} 0 \\ 4e^{-t} \end{bmatrix} + \begin{bmatrix} 2(1 - e^{-t}) \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} 2(1 - e^{-t}) \\ 4e^{-t} \end{bmatrix}. \end{aligned}$$

Answer 6. For:

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix},$$

we have $A^2 = 0$ and:

$$e^{A(t-s)} = \begin{bmatrix} 1 & t-s \\ 0 & 1 \end{bmatrix}.$$

The initial state is zero, so the homogeneous response is zero. Replace t by the integration variable s inside the forcing:

$$\mathbf{f}(s) = \begin{bmatrix} 0 \\ s \end{bmatrix}.$$

Multiply before integrating:

$$\begin{aligned} e^{A(t-s)}\mathbf{f}(s) &= \begin{bmatrix} 1 & t-s \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ s \end{bmatrix} \\ &= \begin{bmatrix} s(t-s) \\ s \end{bmatrix}. \end{aligned}$$

For the first component, expand and integrate:

$$\begin{aligned} \int_0^t s(t-s) ds &= \int_0^t (ts - s^2) ds \\ &= \left[\frac{ts^2}{2} - \frac{s^3}{3} \right]_0^t \\ &= \frac{t^3}{2} - \frac{t^3}{3} \\ &= \frac{t^3}{6}. \end{aligned}$$

For the second component:

$$\int_0^t s ds = \frac{t^2}{2}.$$

Thus:

$$\boxed{\mathbf{x}(t) = \begin{bmatrix} t^3/6 \\ t^2/2 \end{bmatrix}}.$$

The first row is $x'_1 = x_2$; indeed, $(t^3/6)' = t^2/2$.

Answer 7. Use the factored formula with $t_0 = 0$ and $\mathbf{x}(0) = \mathbf{0}$:

$$\mathbf{x}(t) = e^{At} \int_0^t e^{-As} \mathbf{f}(s) ds.$$

For:

$$A = \begin{bmatrix} 1 & 0 \\ 0 & -2 \end{bmatrix},$$

we have:

$$e^{-As} = \begin{bmatrix} e^{-s} & 0 \\ 0 & e^{2s} \end{bmatrix}.$$

Multiply by the forcing at time s :

$$\begin{aligned} e^{-As} \mathbf{f}(s) &= \begin{bmatrix} e^{-s} & 0 \\ 0 & e^{2s} \end{bmatrix} \begin{bmatrix} e^s \\ e^{-2s} \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ 1 \end{bmatrix}. \end{aligned}$$

Integrate:

$$\int_0^t \begin{bmatrix} 1 \\ 1 \end{bmatrix} ds = \begin{bmatrix} t \\ t \end{bmatrix}.$$

Finally:

$$\begin{aligned} \mathbf{x}(t) &= \begin{bmatrix} e^t & 0 \\ 0 & e^{-2t} \end{bmatrix} \begin{bmatrix} t \\ t \end{bmatrix} \\ &= \begin{bmatrix} te^t \\ te^{-2t} \end{bmatrix}. \end{aligned}$$

Both components acquire a factor t because each forcing matches its corresponding homogeneous mode.

Answer 8. Let:

$$A = \begin{bmatrix} 0 & 0 \\ 0 & -1 \end{bmatrix}.$$

Then:

$$e^{At} = \begin{bmatrix} 1 & 0 \\ 0 & e^{-t} \end{bmatrix}, \quad e^{-At} = \begin{bmatrix} 1 & 0 \\ 0 & e^t \end{bmatrix}.$$

Multiply by the forcing:

$$\begin{aligned} e^{-At}\mathbf{f}(t) &= \begin{bmatrix} 1 & 0 \\ 0 & e^t \end{bmatrix} \begin{bmatrix} 2t \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} 2t \\ e^t \end{bmatrix}. \end{aligned}$$

Choose the antiderivative:

$$\int e^{-At}\mathbf{f}(t) dt = \begin{bmatrix} t^2 \\ e^t \end{bmatrix}.$$

Let $\mathbf{c} = (c_1, c_2)^T$. The general formula gives:

$$\begin{aligned} \mathbf{x}(t) &= e^{At} \left[\mathbf{c} + \begin{bmatrix} t^2 \\ e^t \end{bmatrix} \right] \\ &= \begin{bmatrix} 1 & 0 \\ 0 & e^{-t} \end{bmatrix} \begin{bmatrix} c_1 + t^2 \\ c_2 + e^t \end{bmatrix} \\ &= \begin{bmatrix} c_1 + t^2 \\ c_2 e^{-t} + 1 \end{bmatrix}. \end{aligned}$$

Thus the two components are $x_1(t) = c_1 + t^2$ and $x_2(t) = c_2 e^{-t} + 1$.

Worked Answers: Problems 9 To 12

Answer 9. Define:

$$x_1 = y, \quad x_2 = y'.$$

Then:

$$x_1' = x_2.$$

From $y'' + 4y = 0$:

$$y'' = -4y,$$

so:

$$x_2' = -4x_1.$$

The state problem is:

$$\mathbf{x}' = \begin{bmatrix} 0 & 1 \\ -4 & 0 \end{bmatrix} \mathbf{x}, \quad \mathbf{x}(0) = \begin{bmatrix} 0 \\ 3 \end{bmatrix}.$$

For this matrix:

$$e^{At} = \begin{bmatrix} \cos(2t) & \frac{1}{2} \sin(2t) \\ -2 \sin(2t) & \cos(2t) \end{bmatrix}.$$

Apply the initial state:

$$\begin{aligned} \mathbf{x}(t) &= \begin{bmatrix} \cos(2t) & \frac{1}{2} \sin(2t) \\ -2 \sin(2t) & \cos(2t) \end{bmatrix} \begin{bmatrix} 0 \\ 3 \end{bmatrix} \\ &= \begin{bmatrix} \frac{3}{2} \sin(2t) \\ 3 \cos(2t) \end{bmatrix}. \end{aligned}$$

Because $x_1 = y$:

$$y(t) = \frac{3}{2} \sin(2t).$$

Its derivative is $3 \cos(2t)$, which agrees with the second state component.

Answer 10. Define:

$$x_1 = y, \quad x_2 = y'.$$

Then $x'_1 = x_2$. Isolate the second derivative:

$$y'' = 1 - y.$$

Therefore:

$$\mathbf{x}' = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \mathbf{x}(0) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

For:

$$A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix},$$

the transition matrix is:

$$e^{A(t-s)} = \begin{bmatrix} \cos(t-s) & \sin(t-s) \\ -\sin(t-s) & \cos(t-s) \end{bmatrix}.$$

The initial-state response is zero. Multiply the transition matrix by the forcing:

$$e^{A(t-s)} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} \sin(t-s) \\ \cos(t-s) \end{bmatrix}.$$

Integrate the first component. Let $r = t - s$, so $dr = -ds$:

$$\begin{aligned} \int_0^t \sin(t-s) ds &= \int_t^0 \sin r (-dr) \\ &= \int_0^t \sin r dr \\ &= [-\cos r]_0^t \\ &= 1 - \cos t. \end{aligned}$$

Integrate the second component similarly:

$$\begin{aligned} \int_0^t \cos(t-s) ds &= \int_0^t \cos r dr \\ &= \sin t. \end{aligned}$$

Thus:

$$\mathbf{x}(t) = \begin{bmatrix} 1 - \cos t \\ \sin t \end{bmatrix}.$$

Recover the original unknown from $x_1 = y$:

$$\boxed{y(t) = 1 - \cos t}.$$

Answer 11.

Statement 1. The formula omits the initial-time shift. Since the state is specified at $t_0 = 5$, the correct homogeneous solution is:

$$\mathbf{x}(t) = e^{A(t-5)} \mathbf{x}_0.$$

At $t = 5$, this gives $e^{A0} \mathbf{x}_0 = \mathbf{x}_0$.

Statement 2. The multiplication order is reversed. A column vector cannot generally multiply an $n \times n$ matrix from the left. The correct forcing response is:

$$\int_0^t e^{A(t-s)} \mathbf{f}(s) ds.$$

Statement 3. The matrix exponential is defined by a power series involving matrix powers, not by exponentiating entries independently:

$$e^{At} = I + At + \frac{A^2 t^2}{2!} + \dots.$$

Entrywise exponentiation is not valid for a general matrix. For example, the diagonal matrix $A = \text{diag}(a, b)$ has $e^{At} = \text{diag}(e^{at}, e^{bt})$. Its off-diagonal entries remain zero, whereas exponentiating the zero entries individually would incorrectly produce $e^0 = 1$.

Answer 12. Define:

$$\mathbf{X}(s) = \mathcal{L}\{\mathbf{x}(t)\}.$$

The forcing transform is:

$$\mathbf{F}(s) = \begin{bmatrix} 0 \\ \frac{1}{s-1} \end{bmatrix}.$$

Calculate $sI - A$:

$$\begin{aligned} sI - A &= \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \\ &= \begin{bmatrix} s & -1 \\ 2 & s+3 \end{bmatrix}. \end{aligned}$$

The right side is:

$$\begin{aligned} \mathbf{x}_0 + \mathbf{F}(s) &= \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{s-1} \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ \frac{1}{s-1} \end{bmatrix}. \end{aligned}$$

Therefore the explicit resolvent equation is:

$$\boxed{\begin{bmatrix} s & -1 \\ 2 & s+3 \end{bmatrix} \mathbf{X}(s) = \begin{bmatrix} 1 \\ \frac{1}{s-1} \end{bmatrix}}.$$

Calculate the determinant:

$$\begin{aligned} \det(sI - A) &= s(s+3) - (-1)(2) \\ &= s^2 + 3s + 2 \\ &= (s+1)(s+2). \end{aligned}$$

The factors agree with the two homogeneous modes e^{-t} and e^{-2t} .

Chapter 26 Final Summary

For the constant-coefficient matrix initial-value problem:

$$\mathbf{x}'(t) = A\mathbf{x}(t) + \mathbf{f}(t), \quad \mathbf{x}(t_0) = \mathbf{x}_0,$$

the solution is:

$$\mathbf{x}(t) = e^{A(t-t_0)}\mathbf{x}_0 + \int_{t_0}^t e^{A(t-s)}\mathbf{f}(s) ds.$$

The first term carries the initial state. The integral accumulates forcing contributions and carries each one from its input time s to the observation time t .

Without initial conditions:

$$\mathbf{x}(t) = e^{At}\mathbf{c}$$

for a homogeneous system, while a forced general solution is:

$$\mathbf{x}(t) = e^{At} \left[\mathbf{c} + \int e^{-As}\mathbf{f}(s) ds \right].$$